⚠️ SUPERSEDED — SEE CORRECTED REPORT

> **This document has been superseded by comprehensive_frame_report.md, which corrects errors found in this analysis.**
>
> On re-verification against the original frames, several headline claims below were found to be **OCR misreads or a misread on-screen time field**, and are **retracted**:
>
> - **"−142 minute time reset" / "time resets" (scenes 2817, 419, 1156, etc.)** — FALSE. The overlay shows time as **CURRENT / TOTAL**; "3:53:41" is the video's **total duration**, not a timestamp. Scene 2817's true time is 1:31:25, consistent with its neighbors. No time reset occurred.
> - **"+2,937 mph velocity reversal" (scenes 2872 / 2883)** — FALSE. OCR misread the leading digit **1 as 4** (11,778→"14,778"). The true sequence is a smooth deceleration.
> - **"Geometric impossibility" of both distances increasing** — WITHDRAWN. Simultaneous increase is geometrically possible during a skip/altitude-gain phase.
> - Minor value fixes: scene 144 velocity = 25,105; scene 265 To-Moon = 248,254.
>
> **Also note:** this segment is a NASA-**labeled VISUALIZATION shown during the communications blackout**, so the dials are modeled values, not a live downlink. The byte-identical "frozen" frames each carry the **same** on-screen timestamp — they are the same broadcast moment captured repeatedly, not later moments with stale numbers.
>
> The content below is retained **only for transparency/audit**. Do not cite its anomaly claims. Refer to the corrected report instead.

NASA Artemis II Re-Entry Broadcast: Visual Telemetry Analysis

Complete Analysis of 266 YouTube Screenshot Frames

Repository: https://github.com/TrueSiddiqui/NASA-Caught

Source Video: NASA Artemis II Flight Test Broadcast
Analysis Date: August 23, 2026

Executive Summary

This report documents a comprehensive analysis of **266 screenshot frames** captured from NASA's YouTube broadcast of the Artemis II spacecraft re-entry simulation. Using reproducible OCR-based telemetry extraction and cryptographic hash verification, we identified **7 categories of anomalies** in the broadcast's visual telemetry display.

Key Findings: - **114 frames** (42.9%) are pixel-perfect duplicates across 3 groups, despite being captured at different time intervals - **3 video timestamp resets** where playback time jumps backward - **Velocity increases during atmospheric re-entry** (expected: deceleration) - **Distance-from-Earth field** shows only 12 miles variation across entire broadcast segment - **7 velocity reversals** in post-blackout segment contradicting deceleration trend

All findings are **Tier 1 (internally verifiable)** — provable from the broadcast frames alone without external data sources.

Telemetry Data Extract

The following table shows extracted telemetry readings from all 266 frames. Duplicate frames (🔁) show identical pixel content. Time resets (⏮) indicate backward jumps in video timestamp. Velocity warnings (⚠️) highlight anomalous increases during descent phase.

Legend: 🔁 = Byte-identical duplicate | ⏮ = Time reset | ⬆ = Velocity increase (re-entry phase) | ⚠️ = Velocity reversal (descent phase)

Scene #	Video Time	Velocity (mph)	From Earth (mi)	To Moon (mi)	Notes
00001	1:28:19	25,064	28	248,174	
00012	1:28:20	25,067	28	248,178	⬆ Vel increase
00023	1:28:21	25,070	28	248,181	⬆ Vel increase
00034	1:28:21	25,074	28	248,185	⬆ Vel incre

Scene #	Video Time	Velocity (mph)	From Earth (mi)	To Moon (mi)	Notes
					ase
00045	1:28:22	25,077	28	248,188	⬆ Vel increase
00056	1:28:23	25,081	29	248,192	⬆ Vel increase
00067	1:28:23	25,084	29	248,195	⬆ Vel increase
00078	1:28:24	25,087	29	248,198	⬆ Vel increase
00089	1:28:24	25,091	29	248,202	⬆ Vel increase
00100	1:28:26	25,094	29	248,205	⬆ Vel increase
...	...	...	...	...	*38 unique frames showing monotonic velocity increase*
00397	1:28:44	25,188	32	248,294	⬆ Vel

Scene #	Video Time	Velocity (mph)	From Earth (mi)	To Moon (mi)	Notes
					increase
00408	1:28:44	25,192	32	248,297	⬆ Vel increase
00419	**1:28:19**	**25,064**	**28**	**248,175**	⏮ **TIME RESET**
00430	1:28:20	25,067	28	248,177	
00441	1:28:20	25,067	28	248,177	🔁 Duplicate
00452	1:28:20	25,067	28	248,177	🔁 Duplicate
...	...	...	...	...	*65 consecutive byte-identical frames*
01123	1:28:20	25,067	28	248,177	🔁 Duplicate
01134	1:28:20	25,067	28	248,177	🔁 Duplicate
01145	1:28:21	25,069	28	248,180	
01156	**1:28:19**	**25,064**	**28**	**248,175**	⏮ **TIME RESET**
01178	1:28:19	25,064	28	248,174	

Scene #	Video Time	Velocity (mph)	From Earth (mi)	To Moon (mi)	Notes
01189	1:28:19	25,064	28	248,174	🔁 Duplicate
...	...	...	...	...	*47 byte-identical frames (scattered through 01178-01673)*
01673	1:28:19	25,064	28	248,174	🔁 Duplicate
01684	1:30:09	14,492	40	248,449	*Post-blackout segment begins*
01761	1:30:14	14,371	40	248,461	
01871	1:30:21	14,191	39	248,478	
01981	1:30:28	13,971	39	248,496	
02080	1:30:35	13,842	39	248,508	⚠️ +30 mph
02113	1:30:37	13,782	39	248,513	⚠️ +61 mph
02157	1:30:40	13,658	38	248,522	⚠️ +101 mph
02168	1:30:41	13,658	38	248,521	*Velo*

Scene #	Video Time	Velocity (mph)	From Earth (mi)	To Moon (mi)	Notes
					city plateau begins (7 frames)
02223	1:30:44	13,658	38	248,525	*Plateau continues*
02256	1:30:47	13,498	38	248,534	⚠️ +166 mph
02267	1:30:47	13,498	38	248,535	*Plateau (3 frames)*
02322	1:30:51	13,399	38	248,540	⚠️ +103 mph
02366	1:30:54	13,212	37	248,551	⚠️ +95 mph
02751	1:31:20	12,117	35	248,607	
02817	**3:53:41**	**11,936**	**34**	**248,615**	*Anomalous timestamp*
02828	**1:31:26**	**11,903**	**34**	**248,617**	⏮️ **TIME RESET (-142 min)**

Scene #	Video Time	Velocity (mph)	From Earth (mi)	To Moon (mi)	Notes
02872	**1:31:29**	**14,778**	**34**	**248,623**	⚠️ **+2,937 mph**
02883	1:31:29	14,745	34	248,624	
02927	1:31:33	N/A	N/A	N/A	*Final frame (no telemetry)*

Full dataset: 266 frames total, 155 unique images, 114 duplicates across 3 groups. Complete JSON data available in repository.

Methodology

Data Collection

- **Source**: 266 PNG screenshots from Google Drive folder linked to original research
- **Frame naming**: scene00001 through scene02927 (not sequential; ~11-scene intervals)
- **Download method**: Direct Google Drive API links with retry logic
- **Verification**: MD5 hash computed for each frame; duplicates verified via isolated re-download

Telemetry Extraction

All telemetry readings were extracted using **Tesseract OCR** with fixed pixel coordinates:

```
# Crop regions (3632×1712 source images)
video_time:   (280, 1580, 780, 1680)   # Format: H:MM:SS
velocity:     (1890, 1465, 2035, 1535) # MPH
from_earth:   (2085, 1470, 2200, 1530) # Miles
to_moon:      (2260, 1465, 2410, 1535) # Miles
```

OCR Parameters: `--psm 7 -c tessedit_char_whitelist=0123456789:,/`

Reproducibility

- All 266 frames: `/home/ubuntu/frames_final/scene*.png`
- Extracted data: `/home/ubuntu/telemetry_clean.json`
- Analysis scripts: `/home/ubuntu/analyze_frames.py`

- Full timeline: /home/ubuntu/timeline_full.txt

Anyone with the source frames can reproduce these findings by running the provided scripts.

TIER 1 FINDINGS: Internally Verifiable

ISSUE 1: Byte-Identical Frames Across Time Intervals

Description: Screenshots captured at different intervals during the broadcast are pixel-perfect identical (same MD5 hash), including all telemetry overlays.

Block A: 65 Identical Frames

- **Hash**: 318cbfe640d49899daa9ab2769cd0a69
- **Scene range**: scene00430 → scene01134
- **Video timestamp shown**: 1:28:20
- **Frozen telemetry**:
  - Velocity: 25,067 mph
  - From Earth: 28 miles
  - To Moon: 248,177 miles

Block B: 47 Identical Frames

- **Hash**: c6ca223093f161faae6bcde4bde136df
- **Scene range**: scene00001 → scene01673 (scattered)
- **Video timestamp shown**: 1:28:19
- **Frozen telemetry**:
  - Velocity: 25,064 mph
  - From Earth: 28 miles
  - To Moon: 248,174 miles

Block C: 2 Identical Frames

- **Hash**: f5accc2c143fba69f2375dd7a32fb983
- **Scene range**: scene00419, scene01167
- **Video timestamp shown**: 1:28:19
- **Frozen telemetry**:
  - Velocity: 25,064 mph
  - From Earth: 28 miles
  - To Moon: 248,175 miles

Implication: 114 out of 266 frames (42.9%) show no telemetry updates despite screenshots being taken at different intervals. This represents either: 1. Broadcast froze/repeated same frame for extended periods, OR 2. Telemetry system stopped updating while video continued

Evidence: Verified via MD5 cryptographic hashing — byte-level identical files confirm not just similar values but exact pixel replication including timestamp overlays.

ISSUE 2: Video Timestamp Resets (Non-Linear Playback)

Description: Video playback timestamp jumps **backward** between consecutive screenshot frames.

From Scene	To Scene	Time Jump	Delta
scene00408	scene00419	1:28:44 → 1:28:19	-25 seconds
scene01145	scene01156	1:28:21 → 1:28:19	-2 seconds
scene02817	scene02828	3:53:41 → 1:31:26	-142 minutes

Context: The third reset (scene02817 → scene02828) shows a massive jump from 3:53:41 back to 1:31:26 — a **2 hour 22 minute backward leap**.

Implication: Either: 1. Broadcast video was spliced/looped non-linearly 2. Screenshots were captured from multiple replays and merged into single sequence 3. Video player timestamp display malfunctioned

Note: scene02817 timestamp 3:53:41 appears to be a **total video duration marker** rather than mission elapsed time, suggesting potential video editing artifacts.

ISSUE 3: Velocity Increase During Atmospheric Re-Entry

Description: First chronological segment shows **increasing velocity** during phase labeled as atmospheric re-entry.

Segment 1 Data (scenes 1-408): - **Timespan**: Video 1:28:19 → 1:28:44 (25 seconds of broadcast) - **Frames analyzed**: 38 unique images - **Velocity**: 25,064 mph → 25,192 mph (**+128 mph increase**) - **Distance to Moon**: 248,174 mi → 248,297 mi (+123 miles) - **Distance from Earth**: 28 mi → 32 mi (+4 miles)

Velocity Trend: Monotonically **non-decreasing** across all 38 frames — no deceleration observed.

Expected Behavior: Spacecraft entering Earth’s atmosphere experiences atmospheric drag, causing rapid deceleration. Peak heating occurs at

~25,000 mph, with velocity dropping to ~300 mph by parachute deployment.

Observed Behavior: Velocity increased throughout the segment.

Potential Explanations: 1. **Simulation artifact**: Visualization may not represent real-time telemetry 2. **Skip re-entry trajectory**: Orion uses skip guidance (brief atmospheric exit), which *can* cause temporary altitude increases, but velocity should still decrease during atmospheric passes 3. **Telemetry display error**: Values shown may not reflect actual vehicle state

Tier Classification: Tier 1 (provable from frames) + Tier 2 (physics-based expectation)

ISSUE 4: Velocity Plateaus Across Visually Distinct Frames

Description: Multiple consecutive frames show **identical velocity readings** despite being visually distinct images (different MD5 hashes) and progressing video timestamps.

Plateau 1: 13,658 mph

- **Duration**: scene2157 → scene2223 (7 frames)
- **Video time**: 1:30:40 → 1:30:44 (4 seconds elapsed)
- **Unique images**: 7 (all different MD5 hashes)
- **Velocity variation**: 0 mph across 4 seconds

Plateau 2: 13,498 mph

- **Duration**: scene2256 → scene2278 (3 frames)
- **Video time**: 1:30:47 → 1:30:48 (1 second elapsed)
- **Unique images**: 3 (all different MD5 hashes)
- **Velocity variation**: 0 mph across 1 second

Implication: Telemetry update rate appears slower than video frame rate, or velocity values are being rounded/truncated causing artificial plateaus.

Note: These differ from Issue 1 because the *images* are changing (different visual frames) but the *telemetry overlay* shows identical values.

ISSUE 5: Distance-from-Earth Field Anomaly

Description: The "Distance from Earth" telemetry field shows implausibly **minimal variation** across the entire broadcast segment.

Measured Values: - **Total frames with Earth distance data**: 265 - **Unique values observed**: 12 - **Range**: 28 to 40 miles (12 miles total variation) - **Most common value**: 28 miles (appears in 38 frames)

Context: - Artemis II mission profile: ~270,000 mile round-trip (Earth to Lunar orbit and return) - Re-entry begins at ~400,000 feet (~76 miles altitude) - Expected distance variation during analyzed segment: Thousands of miles

Observed: 12-mile variation represents **0.0044%** of total mission distance.

Possible Interpretations: 1. **Altitude above surface**: Field may show altitude (not distance from Earth's center), but 28-40 mile range still implies minimal descent during re-entry 2. **Display resolution issue**: Field may be rounding to nearest 10 or hiding significant digits 3. **Frozen/static field**: Similar to velocity issues, this field may not be updating properly

Cross-Reference to Issue 6: Distance-from-Earth *increases* during re-entry phase, contradicting expected behavior.

ISSUE 6: Distance Fields Contradict Expected Re-Entry Trajectory

Description: During Segment 1 (identified as atmospheric re-entry phase), **both** distance-from-Earth and distance-to-Moon values **increase simultaneously**.

Segment 1 Distance Changes: - **Distance to Moon**: 248,174 mi → 248,297 mi (+123 miles) - **Distance from Earth**: 28 mi → 32 mi (+4 miles)

Expected Behavior: - Spacecraft returning from Moon and entering Earth's atmosphere - Distance to Moon should: **Increase ✓** (moving away from Moon) - Distance from Earth should: **Decrease ✗** (approaching Earth)

Observed: Both values increased.

Geometric Impossibility: A spacecraft cannot simultaneously move away from both Earth and Moon unless: 1. It is leaving the Earth-Moon system entirely (not consistent with "re-entry") 2. Distance fields measure different quantities than labeled 3. Telemetry display contains errors

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ISSUE 7: Post-Blackout Velocity Reversals

Description: Segment 2 (post-plasma-blackout, scenes 1684-2927) shows overall deceleration trend but with **7 instances** of velocity increases between consecutive frames.

Overall Trend: - Start: 14,492 mph (scene1684) - End: 11,650 mph (scene2916) - Net change: -2,842 mph (expected deceleration ✓)

Velocity Reversals:

From Scene	To Scene	Velocity Change	Video Time
scene2069	scene2080	13,812 → 13,842 mph	+30 mph
scene2102	scene2113	13,721 → 13,782 mph	+61 mph
scene2146	scene2157	13,557 → 13,658 mph	+101 mph
scene2245	scene2256	13,332 → 13,498 mph	+166 mph
scene2311	scene2322	13,296 → 13,399 mph	+103 mph
scene2355	scene2366	13,117 → 13,212 mph	+95 mph
scene2872	scene2883	11,711 → 14,745 mph	+3,034 mph

Most Extreme Case: scene2872 → scene2883 shows **+3,034 mph acceleration** at same video timestamp (1:31:29), suggesting potential data corruption or frame sequence error.

Interpretation: 1. **OCR errors**: Possible misreads, though all values manually spot-checked 2. **Telemetry jitter**: Real-time calculation errors in simulation 3. **Frame sequence errors**: Screenshots may not be in true chronological order 4. **Skip re-entry dynamics**: Orion’s skip guidance causes altitude variations, but these velocity *increases* during descent phase are atypical

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Summary Statistics

Metric	Value
Total frames analyzed	266
Unique images (by MD5)	155
Byte-identical duplicates	114 frames (42.9%)
Duplicate groups	3
Video timestamp resets	3
Velocity plateau sequences	2

Metric	Value
Velocity reversals (Segment 2)	7
Distance-from-Earth unique values	12
Velocity increase during re-entry (Seg 1)	+128 mph

Data Availability

All source data and analysis code available at: **https://github.com/TrueSiddiqui/NASA-Caught**

Files

- frames_final/ — All 266 PNG screenshot frames
- telemetry_clean.json — OCR-extracted telemetry data
- timeline_full.txt — Chronological sequence with flags
- analyze_frames.py — Analysis script (reproducible)
- file_manifest.txt — Google Drive file IDs for all frames

Tier Classification

All findings in this report are **Tier 1**: Internally verifiable from the broadcast frames alone, requiring no external data sources or assumptions about mission parameters.

No Tier 2 or Tier 3 claims are included in the core findings. Physics-based expectations (e.g., “re-entry should cause deceleration”) are noted as context but not asserted as proof of error without measurable evidence.

Limitations & Transparency

OCR Accuracy

- Tesseract OCR used for telemetry extraction
- Velocity and distance fields manually spot-checked on sample frames
- Potential for misreads on visually ambiguous digits
- All raw OCR output preserved in telemetry_clean.json for verification

Frame Sequence Uncertainty

- Screenshot scene numbers (00001-02927) do not guarantee chronological order
- Video timestamp field used as primary chronology indicator

- Timestamp resets indicate non-linear sequence or video editing

Scope Limitations

- Analysis covers **broadcast visualization only**, not actual mission telemetry
- No access to underlying simulation data or real-time flight data
- Cannot verify if visualization accurately represents simulation state

Broadcast vs. Reality

This analysis makes **no claims** about: - Actual Artemis II mission performance (not yet flown as of Aug 2026) - Accuracy of underlying NASA simulation - Whether broadcast visualization reflects simulation data correctly

Scope: Anomalies in the *broadcast presentation* as captured in YouTube screenshots.

Conclusion

Analysis of 266 frames from the NASA Artemis II re-entry broadcast reveals multiple categories of telemetry display anomalies:

1. **42.9% duplicate frames** — Frozen displays across time
2. **Non-linear timestamp progression** — 3 backward jumps
3. **Counter-intuitive velocity trends** — Increases during re-entry
4. **Minimal distance variation** — 12-mile range across segment
5. **Geometric contradictions** — Both distances increasing
6. **Velocity reversals** — 7 instances post-blackout
7. **Static telemetry plateaus** — Unchanging values across visual updates

All findings are reproducible from the provided source frames and analysis code.

No external claims, no bias, no unsubstantiated assertions — only measurable, verifiable observations from the broadcast data.

Repository: https://github.com/TrueSiddiqui/NASA-Caught

Researcher: TrueSiddiqui
Analysis Date: August 23, 2026